

StrayCare Platform

Technical Details & System Usability Overview — Draft
AI-Powered Stray Animal Welfare System

1. Technology Stack

Layer	Technology	Details
Backend Framework	FastAPI (Python 3.11)	RESTful API, async routes, auto Swagger docs, CORS middleware
Frontend Framework	React.js 18	SPA with CSS Modules, React Router, custom hooks
Database	SQLite + SQLAlchemy ORM	Relational DB, auto-migration on startup, relationship mapping
Authentication	JWT (OAuth2 Bearer Tokens)	Scoped tokens: user / ngo / admin; bcrypt via passlib
Google Login	Google OAuth 2.0	ID token server-side verification, auto user creation
AI Chatbot	Groq API – LLaMA 3.3-70B	Multi-turn conversation, domain-scoped system prompt
Animal Recognition	TensorFlow.js + MobileNetV2	In-browser inference, no API key, offline after first load
Payment Gateway	Razorpay	Order creation, webhook verification, NGO-linked donations
Geospatial	Haversine Formula	Great-circle distance for smart NGO dispatch scoring
Clustering	Custom K-Means (Python)	Hotspot detection from GPS case coordinates, k=5
Server	Uvicorn (ASGI)	Production-grade async server for FastAPI
File Storage	Local uploads/ directory	Static file serving via FastAPI StaticFiles mount

2. Core System Features

Smart NGO Dispatch

Uses the Haversine formula to calculate real-world GPS distance (km) between a reported case and every verified NGO. Weighted scoring: Proximity 50 pts + Caseload 30 pts + Base 20 pts. Returns a sorted NGO recommendation list for fastest dispatch.

AI Pet Matchmaker

Rule-based compatibility engine scores each available pet against an adopter's lifestyle profile (living space, activity level, presence of children). Score range: 15–99%. Results sorted by compatibility percentage to suggest the best match.

Case Severity Triaging

NLP keyword scoring of case descriptions. Critical keywords (bleeding, seizure, unconscious) add 100 pts; High keywords (injured, poisoned, trapped) add 60 pts; Moderate (limping, sick, malnourished) add 30 pts. Labels: Critical / High / Moderate / Low. NGO queue sorted by severity.

Zone Red Hotspot Map

Custom K-Means clustering (k=5, max 20 iterations) applied to GPS coordinates of reported cases. Cluster radius and risk level mapped to case count: >=10 Critical (800m), >=5 High (500m), >=2 Moderate (300m), 1 Low Zone (150m). Displayed as a color-coded map overlay.

AI Animal Recognition

TensorFlow.js MobileNetV2 runs entirely client-side in the browser. No external API call or key required. Classifies animal species from an uploaded photo using ImageNet-1000 labels mapped to StrayCare categories (Dog, Cat, Bird, Cow, Goat, Other). Model lazy-loads and is cached offline.

Groq AI Chatbot

LLaMA 3.3-70B model served via Groq's OpenAI-compatible API. Supports multi-turn conversation history. Domain-scoped system prompt covers animal first aid, safe rescue guidance, StrayCare app help, and general

animal welfare. Graceful fallback on rate-limit, invalid key, or timeout.

Donor Transparency Dashboard

Public endpoint (/donor/dashboard) exposes month-wise and year-wise donation totals, case stats, and adoption figures without requiring authentication. Builds donor trust through open data.

Donor KYC Verification

OTP-based email and phone verification for donors (dev mode returns code inline; production would send via email/SMS). Verification status tracked: Unverified / Partial / Verified.

3. System Usability — Role-Based Access Matrix

Feature / Capability	Guest	Citizen	NGO	Admin	Donor
Register / Login	✓	✓	✓	✓	✓
Google OAuth Login	✓	✓	—	—	✓
View Pet Listings	✓	✓	✓	✓	✓
Report Stray Case	—	✓	—	—	—
Track Own Cases	—	✓	—	—	—
Accept / Reject Cases	—	—	✓	—	—
Post Case Updates	—	—	✓	—	—
List Pet for Adoption	—	✓	✓	—	—
Submit Adoption Request	—	✓	—	—	✓
Approve Adoption Request	—	—	✓	—	—
Donate to NGO	—	✓	—	—	✓
Donor Verification (KYC)	—	✓	—	—	✓
Leave Feedback / Rating	—	✓	—	—	—
Respond to Feedback	—	—	✓	—	—
NGO Analytics Dashboard	—	—	✓	—	—
Approve / Reject Pet Listings	—	—	✓	—	—
Verify / Deregister NGO	—	—	—	✓	—
View All Platform Feedback	—	—	—	✓	—
Platform Analytics	—	—	—	✓	—
AI Chatbot Access	✓	✓	✓	✓	✓
Animal Recognition (AI)	—	✓	—	—	—
View Hotspot Map	✓	✓	✓	✓	✓
Pet Matchmaker	✓	✓	—	—	—
First Aid Articles	✓	✓	✓	✓	✓

4. Database Schema — Key Tables

Table Name	Columns
users	id, name, email, hashed_password, is_admin
ngos	id, name, email, hashed_password, is_verified, verification_document_url, razorpay_account_id, upi_id

cases	id, description, latitude, longitude, photo_url, status, severity_score, severity_label, owner_id, accepted_by_ngo_id, c
case_updates	id, notes, photo_url, created_at, case_id, ngo_id
pets	id, name, species, breed, age, gender, size, is_vaccinated, location, image_url, video_url, status, ngo_id
adoption_requests	id, pet_id, pet_name, adopter_name, adopter_email, adopter_phone, adopter_address, status, experience, reason, n
donations	id, amount, donor_name, donor_email, payment_id, order_id, status, ngo_id, timestamp
feedback	id, rating, comment, category, ngo_response, user_id, ngo_id, case_id, created_at
user_pet_listings	id, name, species, age, location, description, image_url, status, user_id, created_at
donor_verifications	id, user_id, email_verified, phone_verified, email_code, phone_code, phone_number, verification_status, verified_at

5. Key API Endpoints

Method	Route	Auth	Description
POST	/users/register	—	Citizen registration
POST	/token	—	Citizen login → JWT
POST	/auth/google	—	Google OAuth login, auto-creates user
POST	/ngos/register	—	NGO registration with document upload
POST	/ngo/token	—	NGO login → scoped JWT
POST	/report	Citizen	Report stray case with GPS + photo
GET	/ngo/me/cases	NGO	Severity-sorted case queue for NGO
PUT	/case/{id}/accept	NGO	Accept case and assign to NGO
POST	/cases/{id}/updates	NGO	Post case progress update with photo
GET	/pets	—	List all available pets for adoption
POST	/pets/match	—	AI Pet Matchmaker — scored suggestions
POST	/adoption-requests	—	Submit adoption request for a pet
POST	/chatbot/query	—	Multi-turn Groq AI chatbot
GET	/hotspots	—	K-Means clustered danger zones map
GET	/stats/summary	—	Platform KPIs (cases, adoptions, NGOs)
GET	/donor/dashboard/donations/monthwise	—	Month-wise donation totals
PUT	/admin/ngos/{id}/verify	Admin	Verify NGO and grant platform access
DELETE	/admin/ngos/{id}	Admin	Deregister NGO with reason
POST	/donations/create-order	Citizen	Create Razorpay order for donation
POST	/donations/verify	Citizen	Verify Razorpay payment signature
POST	/feedback	Citizen	Submit star rating + comment for case
PUT	/feedback/{id}/respond	NGO	NGO reply to a citizen review